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|  | <b>Title</b> | <b>Data transmission protocol</b>             |
|                                                                                  | <b>File</b>  | PO60_transmissionprotocol_rev01_20190802.docm |

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## 1. Bluetooth Low energy

### 1.1 General Information

The PO60 is able to transfer stored measurement data via Bluetooth Low energy. The PO60 always works as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet. The newer version is using a 6 Digit PassKey.

### 1.2 Advertising Data

The advertising name is "PO60" and the advertising data of the newer version contains the following information:

| Name                       | Assigned Number | Description                                                                                                                                       | Note |
|----------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Manufacturer Specific Data | 0xFF            | 0x110601<br>The first two bytes 0x0611 are the company ID of "Beurer".<br>The 0x01 is used to identify devices that are using the passkey feature |      |

### 1.3 Services and Characteristics

| Specification Name  | Assigned Number | UUID                                 | Note |
|---------------------|-----------------|--------------------------------------|------|
| Generic Access      | 0x1800          | 00001800-0000-1000-8000-00805f9b34fb |      |
| Generic Attribute   | 0x1801          | 00001801-0000-1000-8000-00805f9b34fb |      |
| Custom Service PO60 | 0xFF12          | 0000FF12-0000-1000-8000-00805f9b34fb |      |
| Device Information  | 0x180A          | 0000180A-0000-1000-8000-00805f9b34fb |      |

#### Generic Access

| Name                                       | Assigned Number | Properties  | Note |
|--------------------------------------------|-----------------|-------------|------|
| Device Name                                | 0x2A00          | Read        |      |
| Appearance                                 | 0x2A01          | Read        |      |
| Peripheral Privacy Flag                    | 0x2A02          | Read, Write |      |
| Reconnection Address                       | 0x2A03          | Write       |      |
| Peripheral Preferred Connection Parameters | 0x2A04          | Read        |      |

#### Generic Attribute

| Name            | Assigned Number | Properties | Note |
|-----------------|-----------------|------------|------|
| Service Changed | 0x2A05          | Indicate   |      |

#### Custom Service

| Name                   | UUID   | Properties       | Note                                                     |
|------------------------|--------|------------------|----------------------------------------------------------|
| Unknown characteristic | 0xFF01 | Write (no resp.) | Send write commands to this characteristic (Data in)     |
| Unknown characteristic | 0xFF02 | Notify           | Provides the requested data via notifications (Data out) |
| Unknown characteristic | 0xFF03 | Read, Write      | (Baud rate)                                              |
| Unknown characteristic | 0xFF04 | Read, Write      | (Link interval)                                          |
| Unknown characteristic | 0xFF05 | Read, Write      | (Pair code)                                              |
| Unknown characteristic | 0xFF06 | Read, Write      | (BT name)                                                |
| Unknown characteristic | 0xFF07 | Read, Write      | (Pair code en)                                           |
| Unknown characteristic | 0xFF08 | Read, Write      | (ADV interval)                                           |
| Unknown characteristic | 0xFF09 | Read, Write      | (TX power)                                               |
| Unknown characteristic | 0xFF0A | Read, Write      | (MCU delay)                                              |
| Unknown characteristic | 0xFFF0 | Write            | (Factory set)                                            |
| Unknown characteristic | 0xFFF1 | Write            | (Reset)                                                  |
| Unknown characteristic | 0xFFF2 | Read             | (FW version)                                             |

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## Device Information

| Name                                                      | Assigned Number | Properties | Note                                       |
|-----------------------------------------------------------|-----------------|------------|--------------------------------------------|
| Manufacturer Name String                                  | 0x2A29          | Read       | Manufacturer Name                          |
| Model Number String                                       | 0x2A24          | Read       | Model Number                               |
| Serial Number String                                      | 0x2A25          | Read       | Serial Number                              |
| Hardware Revision String                                  | 0x2A27          | Read       | Hardware Revision                          |
| Firmware Revision String                                  | 0x2A26          | Read       | Firmware Revision                          |
| Software Revision String                                  | 0x2A28          | Read       |                                            |
| System ID                                                 | 0x2A23          | Read       | 0xF6 8A 1C 00 00 B8 5F D0                  |
| IEEE 11073-20601<br>Regulatory Certification<br>Data List | 0x2A2A          | Read       | 0xFE 00 65 78 7065 72 69 6D 65 6E 74 61 6C |
| PnP ID                                                    | 0x2A50          | Read       | 0x00 0D                                    |

## 1.4 Connection and communication with PO60

A Bluetooth connection by the PO60 is initiated by pressing the on button for several seconds. If no packets are sent for 1000 ms the PO60 stops the Bluetooth connection. In the newer version the advertising is also triggered after each measurement when Bluetooth is enabled on the PO60.

Every packet of the custom service which is send to the PO60 or received from the device has a header byte at the beginning and a checksum byte at the end. The data is between the header and the checksum. The data length is depending on the type of the message. The header is used for identifying what type of message is send or received. The checksum is used to check for transmission errors during the data transfer. The header value is defined in the particular write command or notification in 1.2.2. The checksum is calculated by adding up all the other bytes of the message (including the header) and then select the lowest 7 Bit and write them in the 7 LSB of the checksum byte. The MSB of the checksum byte is always 0.

To communicate with the custom service PO60 write commands are used to give the device information or ask for specific data. The write commands have to be sent to the custom characteristic with the UUID 0xFF01. The device is answering via notifications from the custom characteristic 0xFF02. The other characteristics in the service are not needed for the communication with the device.

## 1.5 Device Version

### Acquire device version – Write

The command for requesting the device version of the PO60.

| Name     | Format | Info                                           | Note |
|----------|--------|------------------------------------------------|------|
| Header   | 1 byte | The header for the device version data is 0x82 |      |
| Checksum | 1 byte | See checksum explanation. Has to be 0x02.      |      |

#### Example:

Sending: [0x82 0x02]

Meaning: 0x82 – Header byte | 0x02 –Checksum

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### Acquire device version – Notify

The response from the PO60 to the acquire device version write command.

| Name             | Format | Info                                                                                                                                                                                                                                                                                                    | Note |
|------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Header           | 1 byte | The header for the device version notification is 0xF2                                                                                                                                                                                                                                                  |      |
| Hardware version | 3 byte | The lowest 7 bit of byte[0] represent the last Version number (A)<br>The lowest 7 bit of byte[1] represent the middle Version number (B)<br>The lowest 7 bit of byte[2] represent the first Version number (C)<br>The Version number would look like: Version C.B.A . The MSB of the bytes are always 0 |      |
| Software version | 3 byte | The lowest 7 bit of byte[0] represent the last Version number (A)<br>The lowest 7 bit of byte[1] represent the middle Version number (B)<br>The lowest 7 bit of byte[2] represent the first Version number (C)<br>The Version number would look like: Version C.B.A . The MSB of the bytes are always 0 |      |
| Checksum         | 1 byte | See checksum explanation in the header and checksum topic                                                                                                                                                                                                                                               |      |

#### Example:

Receiving: [0xF2 00 07 00 01 07 00 01].

Meaning: 0xF2 - header of the notification | [0x00 07 00 - represents the hardware version 0.7.0 | 0x 00 07 01 - software version 0.7.1 | 0x01 - Checksum: You have to sum all the bytes:

$0xF2 + 0x00 + 0x07 + 0x00 + 0x01 + 0x07 + 0x00 = 0x101$

Now you take only the 7 LSB of 0x101 and get 0x01. So the received checksum is correct.

## 1.6 Set time

### Time synchronization – Write

The write command to update the date and time information on the PO60.

| Name         | Format | Info                                                                                                                                                                | Note                                                             |
|--------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Header       | 1 byte | The header for the time synchronization write is 0x83                                                                                                               |                                                                  |
| Year         | 1 byte | The MSB of the byte is always 0. The value of the other 7 bit + the value of 2000 represent the year                                                                |                                                                  |
| Month        | 1 byte | The four most significant bit of the byte are always 0. The other 4 bit represent the month                                                                         |                                                                  |
| Day          | 1 byte | The three most significant bit of the byte are always 0. The other 5 bit represent the days                                                                         |                                                                  |
| Hour         | 1 byte | The three most significant bit of the byte are always 0. The other 5 bit represent the hours                                                                        |                                                                  |
| Minute       | 1 byte | The two most significant bit of the byte are always 0. The other 6 bit represent the minutes                                                                        |                                                                  |
| Second       | 1 byte | The two most significant bit of the byte are always 0. The other 6 bit represent the seconds                                                                        |                                                                  |
| Milliseconds | 2 byte | The MSB of both bytes are always 0. The other 14 bit represent the value in milliseconds. The MSB 7 bit are in the second byte and LSB 7 bit are in the first byte. | Note: Not sure if it works this way. Please set them to 0x00 00. |
| Checksum     | 1 byte | See checksum explanation in the header and checksum topic                                                                                                           |                                                                  |

#### Example:

Sending: [0x 83 10 04 1A 0A 2A 1E 52 03 58]

Meaning: 0x83 – header byte | 0x10 - Year, 2016 | 0x04 - Month with the value 4 | 0x1A - Day with the value 26 | 0x0A - Hour with the value of 10 | 0x2A - Minutes with the value of 42 | 0x1E - Seconds with the value of 30 | 0x03 52 - Milliseconds with the value of 466 | 0x58 - Checksum

### Time synchronization – Notify

The response of the PO60 to the time synchronization write command.

| Name   | Format | Info                                                   | Note |
|--------|--------|--------------------------------------------------------|------|
| Header | 1 byte | The header for the time synchronization notify is 0xF3 |      |

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| Verification | 1 byte | If the value is 0 the time synchronization was successful. If the value is not 0 is failed. |  |
| Checksum     | 1 byte | See checksum explanation in the header and checksum topic                                   |  |

**Example:**

Receiving: [0xF3 0x00 0x73]

Meaning: 0xF3 – Header byte | 0x00 – synchronization was successful | 0x73 –Checksum

## 1.7 Acquire device time

### Acquire device time – Write

Command to request the current time information on the PO60.

| Name     | Format | Info                                                                      | Note |
|----------|--------|---------------------------------------------------------------------------|------|
| Header   | 1 byte | The header for acquire device time write data is 0x89                     |      |
| Checksum | 1 byte | See checksum explanation in the header and checksum topic. Has to be 0x09 |      |

**Example:**

Sending: [0x89 0x09]

Meaning: 0x89 – Header byte | 0x09 –Checksum

### Acquire device time – Notify

The answer of the PO60 to the acquire device time write command.

| Name         | Format | Info                                                                                                                                                                | Note                                                           |
|--------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Header       | 1 byte | The header for acquire device time notify data is 0xF9                                                                                                              |                                                                |
| Year         | 1 byte | The MSB of the byte is always 0. The other 7 bit + the value of 2000 represent the year                                                                             |                                                                |
| Month        | 1 byte | The four most significant bit of the byte are always 0. The other 4 bit represent the month                                                                         |                                                                |
| Day          | 1 byte | The three most significant bit of the byte are always 0. The other 5 bit represent the days                                                                         |                                                                |
| Hour         | 1 byte | The three most significant bit of the byte are always 0. The other 5 bit represent the hours                                                                        |                                                                |
| Minute       | 1 byte | The two most significant bit of the byte are always 0. The other 6 bit represent the minutes                                                                        |                                                                |
| Second       | 1 byte | The two most significant bit of the byte are always 0. The other 6 bit represent the seconds                                                                        |                                                                |
| Milliseconds | 2 byte | The MSB of both bytes are always 0. The other 14 bit represent the value of milliseconds. The MSB 7 bit are in the second byte and LSB 7 bit are in the first byte. | Note: Not sure if it works this way. Please ignore this value. |
| Checksum     | 1 byte | See checksum explanation in the header and checksum topic                                                                                                           |                                                                |

**Example:**

Receiving: [0x F9 10 04 1A 0A 2A 25 63 00 63]

Meaning: 0xF9 – header byte | 0x10 - Year, 2016 | 0x04 - Month with the value 4 | 0x1A - Day with the value 26 | 0x0A - Hour with the value of 10 | 0x2A - Minutes with the value of 42 | 0x25 - Seconds with the value of 37 | 0x00 63 - Milliseconds with the value of 99 | 0x63 - Checksum

## 1.8 Acquire data storage information

### Acquire data storage information – Write

The command to request information about the stored data in the PO60. The data itself is requested by another command.

| Name      | Format | Info                                                                                                                                                 | Note |
|-----------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Header    | 1 byte | The header for acquire data storage information write is 0x90                                                                                        |      |
| Data type | 1 byte | Declares for which data type information will be provided:<br>If the value is 5 : SpO2 and PR data with stored time period and average value (times) |      |
| Checksum  | 1 byte | See checksum explanation in the header and checksum topic                                                                                            |      |

**Example:**

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Sending: [0x90 0x05 0x15]

Meaning: 0x90 – Header byte | 0x05 – SpO2 and PR data are requested | 0x15 –Checksum

### Acquire data storage information – Notify

The information about the stored device from the PO60.

| Name           | Format | Info                                                                                                                        | Note                                                   |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Header         | 1 byte | The header for acquire data storage information notify is 0xE0                                                              |                                                        |
| Data type      | 1 byte | The data type which was requested in the write command. See "Acquire data storage information – Write" for more information |                                                        |
| Total capacity | 2 byte | The total capacity of the stored data                                                                                       |                                                        |
| Unread data    | 2 byte | The capacity of non-uploaded data                                                                                           | Didn't work on tests. All data always shown as unread. |
| Checksum       | 1 byte | See checksum explanation in the header and checksum topic                                                                   |                                                        |

#### Example:

Receiving: [0xE0 05 08 00 08 00 75]

Meaning: 0xE0 – header byte | 0x05 - requested data were SpO2 and PR data with stored time and average values|| 0x00 08 - 8 measurements of this type are stored in the PO60 | 0x00 08 - 8 of these measurements are unread (note: if a measurement has been read and confirmed the value still shows all measurements as unread) | 0x75 - Checksum

## 1.9 Acquire measurement data

### Acquire measurement data – Write

The command to request the stored measurement data. The data will be send in groups of 10 measurements per request.

| Name         | Format | Info                                                                                                                                                                                  | Note |
|--------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Header       | 1 byte | The header for acquire measurement data notify is 0x99                                                                                                                                |      |
| Data command | 1 byte | value 0: start<br>value 1: continue with next group<br>value 2: resend last group<br>value 126: all data received, save the record<br>value 127: all data received, delete the record |      |
| Checksum     | 1 byte | See checksum explanation in the header and checksum topic                                                                                                                             |      |

#### Example:

Sending: [0x99 0x00 0x19]

Meaning: 0x99 – Header byte | 0x00 – Start the transmission | 0x19 –Checksum

### Acquire measurement data – Notify

The structure of one measurement data, send by the PO60 in response to the “acquire measurement data” write command. There will be send up to ten measurements. The data of one measurement is greater than 20 bytes. So the data does not fit into one notification. The rest of the measurement data will be send with the next notification. The header is unique because it is the only byte where the MSB is 1. That means the start of a new measurement can be identified by the header.

| Name                              | Format | Info                                                                                                                                                      | Note |
|-----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Header                            | 1 byte | The header for acquire measurement data notify is 0xE9                                                                                                    |      |
| Packet number and last packet bit | 1 byte | The 4 LSB are the packet number (0-9). Bit 6 is set to 1 if the packet is the last one on the device.                                                     |      |
| Start year                        | 1 byte | This value represents the starting year of the measurement. The MSB of the byte is always 0. The other 7 bit + the value of 2000 represent the year.      |      |
| Start month                       | 1 byte | This value represents the starting month of the measurement. The four most significant bit of the byte are always 0. The other 4 bit represent the month. |      |
| Start day                         | 1 byte | This value represents the starting day of the measurement. The three most significant bit of the byte are always 0. The other 5 bit represent the days.   |      |
| Start hour                        | 1 byte | This value represents the starting hour of the measurement. The three most significant bit of the byte are always 0. The other 5 bit represent the hours. |      |

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|---------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Start minute                    | 1 byte | This value represents the starting minute of the measurement. The two most significant bit of the byte are always 0. The other 6 bit represent the minutes.                                                                                                           |  |
| Start second                    | 1 byte | This value represents the starting second of the measurement. The two most significant bit of the byte are always 0. The other 6 bit represent the seconds.                                                                                                           |  |
| End year                        | 1 byte | This value represents the ending year of the measurement. The MSB of the byte is always 0. The other 7 bit + the value of 2000 represent the year.                                                                                                                    |  |
| End month                       | 1 byte | This value represents the ending month of the measurement. The four most significant bit of the byte are always 0. The other 4 bit represent the month.                                                                                                               |  |
| End day                         | 1 byte | This value represents the ending day of the measurement. The three most significant bit of the byte are always 0. The other 5 bit represent the days.                                                                                                                 |  |
| End hour                        | 1 byte | This value represents the ending hour of the measurement. The three most significant bit of the byte are always 0. The other 5 bit represent the hours.                                                                                                               |  |
| End minute                      | 1 byte | This value represents the ending minute of the measurement. The two most significant bit of the byte are always 0. The other 6 bit represent the minutes.                                                                                                             |  |
| End second                      | 1 byte | This value represents the ending second of the measurement. The two most significant bit of the byte are always 0. The other 6 bit represent the seconds.                                                                                                             |  |
| Storage time period and PR bits | 1 byte | The two MSB of this byte are always 0. Bit 5 represents the MSB of the PR max value. Bit 4 represents the MSB of the PR min value. Bit 3 represents the MSB of the PR avg value. The lowest 3 Bits of the byte represents the 3 MSB of the storage time period value. |  |
| Storage time period             | 2 byte | The MSB of the two bytes are always 0. The other 14 Bit are representing the lower Bits of the storage time period.                                                                                                                                                   |  |
| SpO2 max                        | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the SpO2 maximum value.                                                                                                                                                                                  |  |
| SpO2 min                        | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the SpO2 minimum value.                                                                                                                                                                                  |  |
| SpO2 avg                        | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the SpO2 average value.                                                                                                                                                                                  |  |
| PR max                          | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the LSB of the PR maximum value. The MSB is stored in the "storage time period and PR bits" byte                                                                                                         |  |
| PR min                          | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the LSB of the PR minimum value. The MSB is stored in the "storage time period and PR bits" byte.                                                                                                        |  |
| PR avg                          | 1 byte | The MSB of this byte is always 0. The other 7 Bit represents the LSB of the PR average value. The MSB is stored in the "storage time period and PR bits" byte.                                                                                                        |  |
| Checksum                        | 1 byte | See checksum explanation in the header and checksum topic                                                                                                                                                                                                             |  |

#### Example:

**Receiving:** [0x E9 00 0F 0B 06 0D 14 25 0F 0B 06 0D 14 3A 00 00 16 60 60 60] [0x 6F 6E 6E 4B E9 01 01 01 01 00 00 20 01 01 01 00 00 2F 00 00]

**Meaning:** 0xE9 - Header | 0x00 - packet number | 0x0F – Year 2015 when the measurement started | 0x0B - November and is the starting month of the measurement | 0x06 - measurement started 6<sup>th</sup> day | 0x0D - measurement started on the 13<sup>th</sup> hour | 0x14 - measurement started on the 20<sup>th</sup> minute | 0x25 - measurement started on the 37<sup>th</sup> second | 0x0F - 2015 is the year the measurement ends | 0x0B - end month November | 0x06 - end day 6 | 0x0D - end hour 13 | 0x14 - end minute 20 | 0x3A - end second is 58 | 0x00 - The MSB of all the PR values is 0 and the three MSB of the storage time period are also 0 | 0x00 16 - storage time period is 22 | 0x60 - maximum SpO2 value is 96 | 0x60 - minimum SpO2 value is 96 | 0x60 - average SpO2 value is 96 | [the next bytes are in the next notification] 0x6F - maximum PR from 111 | 0x6E - minimum PR from 110 | 0x6E - average PR from 110 | 0x4B - checksum of the measurement | 0xE9 - Next measurement data is starting...

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| <b>HISTORY</b>  | Changes to the previous document will be <b>marked in red</b> and documented below.<br>Before updating the document, all previous red markings have to be set back to black colour. |                    | <b>Issued/<br/>changed by</b> |
| <b>Revision</b> | <b>Date</b>                                                                                                                                                                         | <b>Description</b> |                               |
| 01              | 2019-08-02                                                                                                                                                                          | Initial version    | MIMA                          |